higher education & training

Department:
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REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

FLUID MECHANICS N6

(8190216)

26 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages and a formula sheet of 2 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
FLUID MECHANICS N6
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Round off final answers to THREE decimals where applicable.
5. Diagrams and sketches in this question paper are NOT drawn to scale.
6. Use g = 9,81 m/s^2.
7. Write neatly and legibly.

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QUESTION 1

1.1 Define the term *turbulent flow*. (2)

1.2 What is *hydraulic mean depth*? (1)

1.3 FIGURE 1 shows water flowing through a pipeline which contracts from 405 mm in diameter at point P to 285 mm in diameter at point Q. It then forks with one branch 255 mm in diameter discharging at S. The other branch with a diameter of 105 mm discharges at R. The velocity at P is 1,8 m/s and the velocity at S is 3,6 m/s.

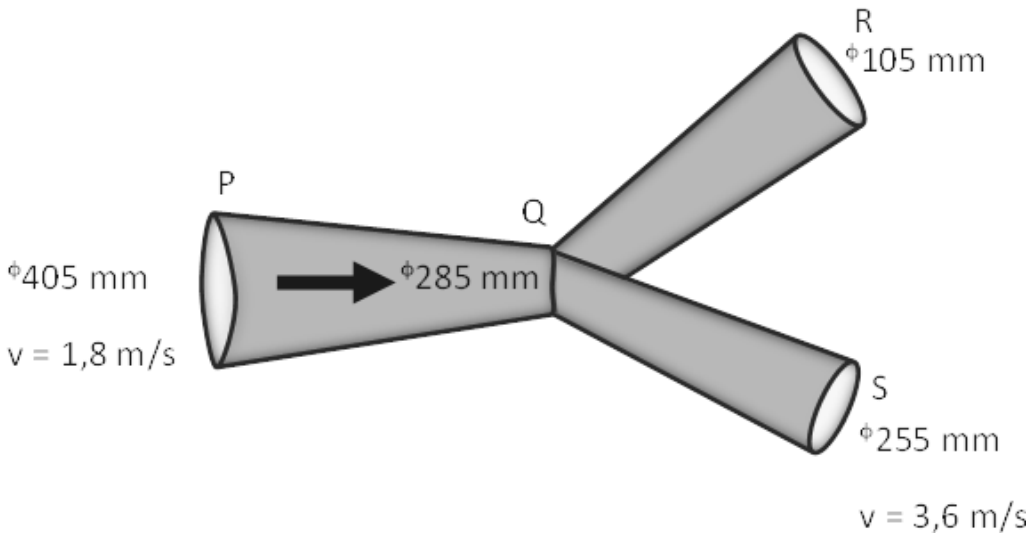


FIGURE 1

Calculate:

1.3.1 The discharge at S in m^3/s (2)

1.3.2 The discharge at R in m^3/s (4)

1.3.3 The velocity at R (2)

1.3.4 The velocity at Q (2)

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1.4 A pipeline conveys oil with a relative density of 0,89 at a rate of 165 m^3/h as shown in FIGURE 2. The pipe tapers uniformly from a diameter of 125 mm to 80 mm. The centre line at the entry is 1,1 m below the discharge end.

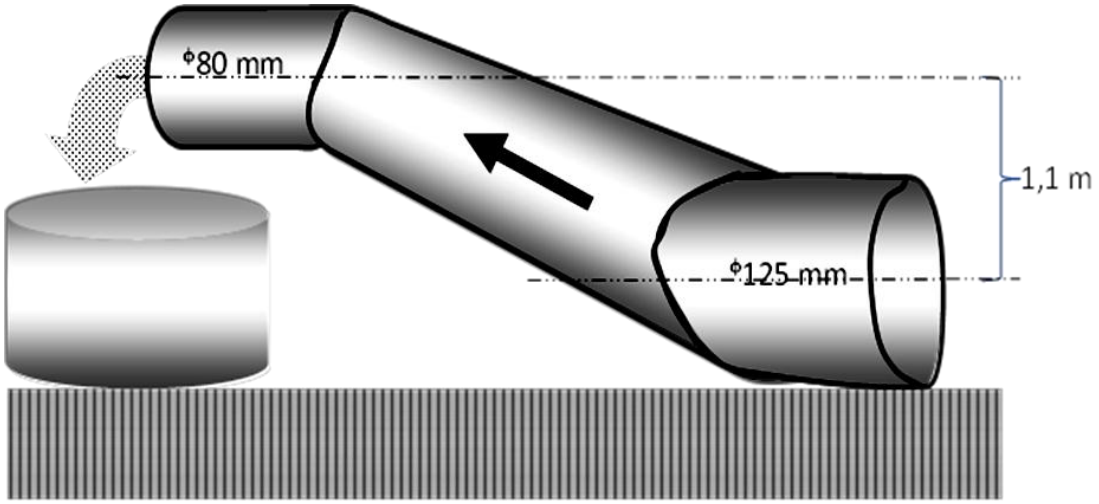


FIGURE 2

Neglect the losses and determine the outlet pressure if the inlet pressure is 160 kPa. (8)

[21]

QUESTION 2

2.1 The cross section of an open rectangular-shaped channel is full of water as shown in FIGURE 3. The gradient of the channel is 1 in 2 550 and C in the Chézy formula is 60. The maximum depth of the channel is 2 500 mm.

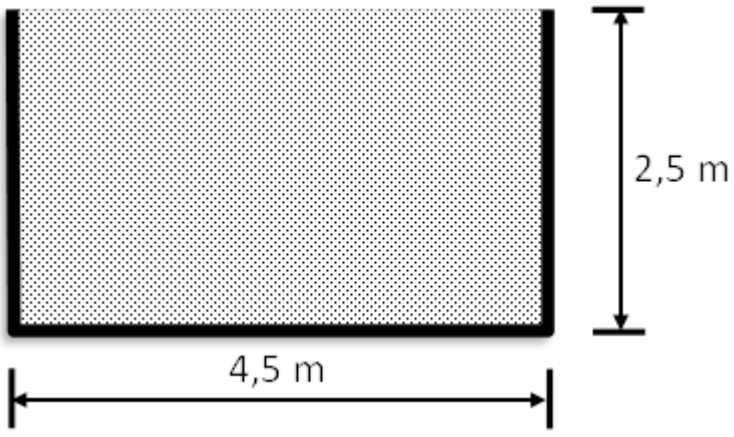


FIGURE 3

Calculate the discharge in m^3/s. (9)

2.2 A tank 2,1 m high standing on the ground is kept full of water as shown in FIGURE 4. There is an orifice at a depth h metres below the surface.

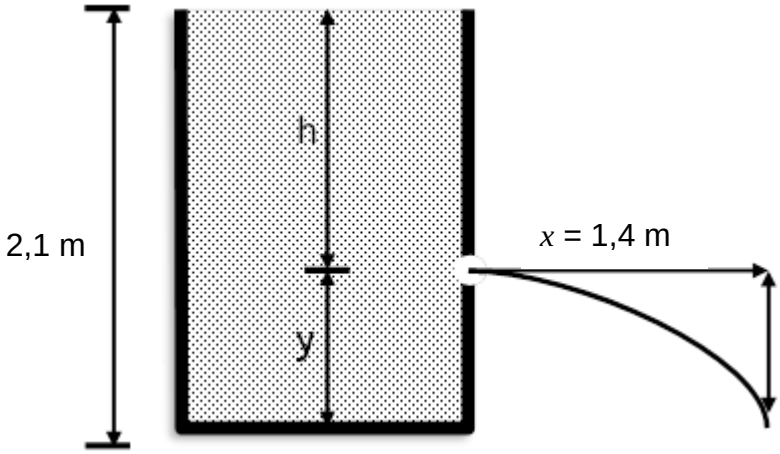


FIGURE 4

Calculate the value of h so that a jet of water from the orifice will strike the ground at a maximum distance of 1,4 m from the tank. The coefficient of velocity is 0,9. (10)
[19]

QUESTION 3

3.1 A single-acting, single-cylinder plunger pump operating at 1,7 r/sec has a diameter of 350 mm and a stroke of 550 mm. The delivery pipe has a diameter of 160 mm, is 40 m long and has a pipe-friction factor of 0,0067.

Determine the power saved to overcome friction in the delivery pipe if a large air vessel is fitted in the pipe close to the pump cylinder. (12)

3.2 A double-acting reciprocating pump has a piston with a diameter of 0,5 m and a stroke of 525 mm. The diameter of the pipe is 130 mm and the water is lifted vertically through a height of 14 m. The pump speed is 65 r/min and the actual quantity of water delivered is 18,4 ℓ/s.

Calculate:

3.2.1 The percentage slip in the pump (4)

3.2.2 The coefficient of discharge (2)

3.2.3 The theoretical power to drive the pump (2)

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3.3 FIGURE 5 shows an indicator diagram.

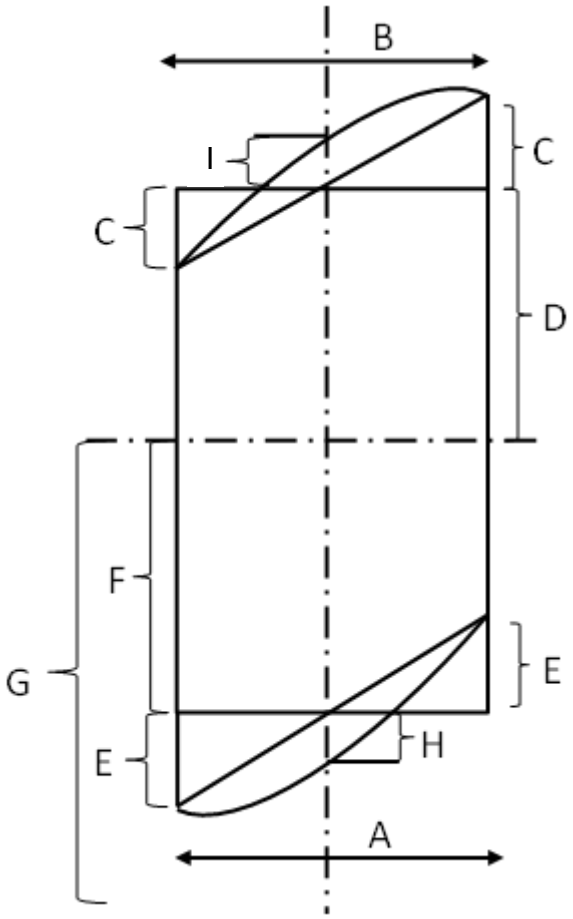


FIGURE 5

Label the diagram by writing only the answer next to the letter (A–I) in the ANSWER BOOK. (9 × 1) (9)

3.4 Give the pressure-head formula for each of the following strokes:

3.4.1 The beginning of the suction stroke

3.4.2 The middle of the delivery stroke

3.4.3 The end of the suction stroke

(3 × 1) (3)

3.5 A ventilation system requires air to be circulated at 654 m^3/min. The total resistance of the circuit together with dynamic pressure has been calculated at 50 mm. A water-gauge fan is available but from its characteristics it is found that at a speed of 500 r/min it will deliver 620 m^3/min at a maximum pressure of 45 mm water gauge.

Determine whether this fan can be used. (5)

[37]

QUESTION 4

4.1 Water is supplied to an axial flow turbine under a total head of 45 m. The mean diameter of the runner is 3 m and it rotates at 150 r/min. Water leaves the guide vanes at 30° to the direction of the runner rotation. 7% of the total head is lost in the casing and the guide vanes and relative velocity is reduced by 8% due to friction in the runner which produces a velocity of whirl at the inlet and outlet of 20 m/s and 5 m/s respectively.

Determine the blade angle at the inlet (at mean radius) and the hydraulic efficiency of the turbine. (14)

4.2 Calculate the power and hydraulic efficiency of a Pelton wheel supplied with water under a head of 60 m at a rate of 43 m^3/min. The bucket deflects a jet of water through an angle of 162° and the mean bucket speed is 17 m/s. (9)

[23]

TOTAL: 100

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FORMULA SHEET

Any applicable formula may be used.

$$Z_1 + \frac{P_{r1}}{\rho g} + \frac{V_1^2}{2g} = Z_2 + \frac{P_{r2}}{\rho g} + \frac{V_2^2}{2g} + h_L$$

$$hf = \frac{4\,fLV^2}{2gd} \qquad hs = \frac{kV^2}{2g}$$

$$hs = \frac{(V_1 - V_2)^2}{2g} \qquad hs = \frac{V^2}{2g} \times \left(\frac{1}{C_c} - 1\right)^2$$

$$Q = A.C\sqrt{mi}$$

$$Q = 1{,}84\ (L - 0{,}1\ n.H)\ H^{1,5}$$

$$Q = \frac{2}{3}\,Cd\sqrt{2g} \times L \times H^{1,5}$$

$$Q = \frac{8}{15}\,Cd\sqrt{2g} \times \tan\frac{\theta}{2} \times H^{2,5}$$

$$Q = \frac{2}{3}Cd\sqrt{2g}\ H^{1,5}\left(L + \frac{4}{5}\tan\frac{\theta}{2} \times H\right)$$

$$Q = \frac{ALSEN}{60}$$

$$Ha = \frac{L}{g} \times \frac{D^2}{d^2} \times \omega^2 \times r \times \cos\theta$$

$$hf = \frac{4\,fL}{2gd} \times \left[\frac{D^2}{d^2} \times \omega \times r\right]^2$$

$$hf = \frac{4\,fL}{2gd} \times \left[\frac{D^2}{d^2} \times \frac{\omega r}{\pi}\right]^2$$

$$\frac{Q_1}{Q_2} = \frac{N_1}{N_2}$$

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$$\frac{P_{r1}}{P_{r2}} = \left(\frac{N_1}{N_2}\right)^2$$

$$\frac{kW_1}{kW_2} = \left(\frac{N_1}{N_2}\right)^3$$

$$\frac{Q_1}{Q_2} = \left(\frac{D_1}{D_2}\right)^3$$

$$\frac{P_{r1}}{P_{r2}} = \left(\frac{D_1}{D_2}\right)^2$$

$$\frac{kW_1}{kW_2} = \left(\frac{D_1}{D_2}\right)^5$$

$$\frac{P_{r1}}{P_{r2}} = \frac{\rho_1}{\rho_2}$$

$$\frac{kW_1}{kW_2} = \frac{1}{\rho}$$

$$\frac{H_1}{H_2} = \left(\frac{Q_1}{Q_2}\right)^2$$

$$\frac{H_1}{H_2} = \left(\frac{N_1}{N_2}\right)^2 \;;\; \frac{w.g._1}{w.g._2} = \left(\frac{N_1}{N_2}\right)^2$$

$$\frac{H_1}{H_2} = \frac{L_1}{L_2}$$

$$\frac{W_1}{W_2} = \left(\frac{D_1}{D_2}\right)^2$$

$$\frac{N_1^2 D_1^2}{gh_1} = \frac{N_2^2 D_2^2}{gh_2}$$

$$P_r = \frac{kSV^2}{a}$$

$$P = \rho \times g \times Q \times w.g.$$

$$P = \rho \times Q \times u(v - u)\,[1 + n \cos(180° - y)]$$

$$\eta = \frac{u}{gh}\,(v - u)\,[1 + n \cos(180° - y)] \times 100$$

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